

PLC Based Monitoring & Protection of 3-Phase Induction Motors against Various Abnormal Conditions

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Abstract – In this manuscript, low-cost and efficient protection scheme for three phase induction motors is designed against the most commonly occurring abnormal conditions – over current, over voltage and over temperature. This protection scheme makes use of the modern control equipments used in the industrial processes. The brain of the scheme – Programmable Logic Controller (PLC) – continuously monitors the different parameters of an induction motor and gives an alert message on Human Machine Interface (HMI) about any abnormal condition detected, and trips the induction motor if no troubleshooting action is taken within defined time. The design has an additional feature of automatically start of induction motor when fault is removed which mitigates the need of a recloser. The design has been tested on different three phase induction motors of different parameters available in laboratory. Keeping operating time in consideration the design replaces the old mechanical relays with modern solid-state relays, and the same relay has been used for signaling for all abnormal conditions which again makes this design an economically low-cost.

Keywords – *Induction Motors, Programmable Logic Controllers, Human Machine Interface, Fault Detection.*

I. INTRODUCTION

Three phase induction motors are the heart of industries. Every industrial process is associated with induction machines either directly or indirectly. Its robustness and simple construction has led to vast industrial applications [1]. When we see importance of induction motors in industries the question of its protection and monitoring arises. Industrial induction motors are used in different processes due to which they may exhibit abnormal conditions like over voltage, over current, over temperature and many others. If these faults are not controlled in the defined time span, it can completely destroy the induction motor. The conventional methods of protection of induction motors uses mechanical relays and microcontroller and other discrete components that led to time delay and other processing problems which could result in damage to the costly industrial motors. Thus when it comes to huge industrial control units the modern method of protection plays an important role.

There are various different techniques which have been used to cope with different mechanical and electrical faults in Induction Motor. Motor Current Signature Analysis (MCSA) is one of the techniques used to overcome the unbalanced supply problem in three phases of motor [11] – [14]. In this techniques, the spectral analysis results of three phase stator current are used to find out the fault in induction motor. To extract the three-phase stator current from signal to the power distribution in frequency domain is known as power spectral density (PSD) [2]. In addition to signature analysis method, MLP Neural Network architecture is another method used to determine the under voltage fault condition [3].

This paper discusses modern protection methodology where complete protection is done through the programmable logic controller (PLC). In this method of protection, the online monitoring of induction motor is done and all the necessary electrical parameters – voltage, current and temperature are monitored. If the parameters are bounded which mean all the parameters are within their normal operating range, the PLC will continuously be allowing the induction motor to be connected with the three phase supply. However, if there is any disturbance found, PLC will trip the induction motor by giving a tripping-signal to magnetic contactor and relay as per the programmed conditions.

The uniqueness of this paper is that it proposes the protection scheme using solid state relays coupled with PLC and magnetic contactors. Hence the drawbacks associated with mechanical relays are eradicated. HMI designed specifically to meet the design criteria of the protection schemes makes statistics and type of abnormal condition immediately available for the operator.

I. PLC AS A SYSTEM CONTROLLER

Programmable Logic Controllers are essentially a computing device, also known as programmable controllers. They are widely used in industrial applications, specifically in automation. The PLC monitors input signal, processes the data received and then makes decisions based on the stored program in the CPU. It then sends the command to the slaves to take necessary actions as shown in Fig. 1.

The controller consists of a CPU (Central Processing Unit), Digital input (DI) and output modules, analog input (AI) and

output modules, and other customized modules [4]. The input of the PLC takes many different types of digital or analog signals, which are fed to PLC through different sensors. Once inputs are given, PLC converts those digital signals into logic signals that can be used by the CPU. After that CPU decides the output status, either it should be logical high or logical low as per the condition set by the programmer in ladder logic.

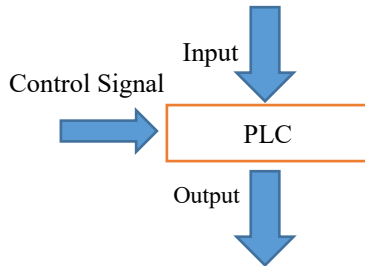


Fig. 1. PLC depicted as Control Unit

In this design of induction motor protection, we are using both applications of digital and analog inputs of PLC. The over voltage abnormal conditions is monitored by converting the analog three phase signal to digital signal and each single phase is fed to the digital input of PLC. However, for over temperature and under voltage abnormal conditions, analog inputs of PLC are used. In both cases, the inputs are continuously monitored and processed by the CPU of PLC and output is activated according to the set condition. Thus CPU is one of the main components of this controller.

Another most essential part of a PLC is HMI (Human Machine Interface). It allows the operator to monitor process measurements such as voltage, current, and speed etc. on a screen and allows user to input parameters to configure a device or a machine. In this project we have used KTP 600 Basic PN HMI which is shown in Fig.2. Following is the basic description of this HMI:

- It is a 6 inch screen with 6 tactile keys
- Color (TFT, 256 colors) or Mono (STN, gray scales)
- 115.2 mm x 86.4 mm (5.7")
- Portrait or landscape
- Resolution: 320 x 240



Fig. 2. KTP 600 Basic PN HMI

II. HARWARE DESCRIPTION FOR OVER VOLTAGE CONDITION

The hardware consists of induction motor, potential transformers (PTs), solid state relay, magnetic contactor and obviously PLC. The protection against over voltage condition for our scheme is achieved by using the digital inputs of controller.

The three phase supply from three phase source is fed to the inputs of three PTs. The output of each PT is stepped-down voltage level from 220 V to 24 V. The output terminals of PTs are connected to individual three rectifier circuits, where each single phase of three phase supply is rectified and then fed to the input terminals of the PLC. The main benefit of using three different rectifier circuit is that, if over voltage occurs in any of the three phases the motor is disconnected from the supply. The rectified input is given to the PLC where it is monitored for each single instant. At the output terminal of the PLC, a solid-state relay is connected and relay is further connected with a three phase magnetic contactor, which triggers on/off the three phase supply of the motor. Thus forming a complete loop. Under normal condition, the output of PLC is low and no tripping signal to solid state relay, which makes magnetic contactor contacts closed, thus making motor to run. While under the abnormal condition, the output of PLC is high which triggers relay and relay further switches the contactor contacts open and thus three phase supply is cut off and motor is safely disconnected from the over voltage condition.

III. HARWARE DESCRIPTION FOR UNDER VOLTAGE CONDITION

Due to change at supply end, over voltage and under voltage occurs. "National Electrical Manufacturer Association (NEMA)", "Institute of Electrical and Electronics Engineers (IEEE)" and "International Electro technical Commission (IEC)" given statement that if unbalanced voltage supply changes at 5% deviation from the standard or normal voltage level, then it will cause higher winding temperature in induction motor [1]. So, this is necessary to protect induction motor from under voltage as well.

In this section, we aim to discuss the under voltage protection scheme for an induction motor. The analog input of PLC is utilized to detect the abnormal voltage. The design has similar structure to over voltage design structure except that analog inputs of controller are used in this case instead of digital input which has only two states of 1 (on) and 0 (off) [8]. The advantage of using analog inputs is that the real time voltage level is continuously monitored, and the output is not bounded to two states: 0 and 1. Under normal operating voltage level, the contacts of magnetic contactor are closed making the circuit closed and keep motor running. The under voltage condition is detected by the controller which sends tripping signal to disconnect the induction motor if no immediate action is taken within time set in controller programming [9].

A. Over and Under Voltage level calculation

Three phase induction motors require balanced three phase voltage conditions. Any phase with negative sequence voltages can cause unbalanced voltage and ultimately rise the current level and severely damage the motor [15]. According to National Electrical Manufacturers Association NEMA voltage unbalance is defined as:

$$V_{unbalance} = \frac{\Delta V_{max}}{V_{avg}} \times 100\%$$

Where ΔV_{max} is the maximum change/deviation of voltages from the average of three phase voltages, while V_{avg} is the average of three phase voltages.

IV. HARWARE DESCRIPTION FOR OVER TEMPERATURE CONDITION

Over temperature or temperature rise is the rise in temperature in the winding of an induction motor when it is operated at full speed. The percentage rise in temperature in induction motor is given by following equation [5].

$$\%(\text{Temp rise}) = 2 * (\text{Voltage unbalance})^2$$

Over-temperature is the most severe condition in induction motors, which initially damages insulation of motor winding and can cause to completely burn out winding. According to IEEE 841 standard for petroleum & chemical industry, the bearing temperature at rated load shouldn't exceed 45 degree centigrade [6]. Winding temperature will increase by 25 % in the phase having the highest current if there is a voltage unbalance of 3.5 % per phase [10]. The previous temperature monitoring technique like conditioning monitoring required both temperature sensor and complex embedded circuit which made the protection scheme more complex and costly, it is one of the great disadvantages [6]. However in our protection scheme against over temperature, only a temperature sensor (LM35) is used.

A. LM 35 Temperature Sensor

LM 35 is a temperature sensor which gives output voltage linearly proportional to Centigrade temperature. It is very advantageous over linear temperature sensors, In order to obtain convenient centigrade scaling, there is no need to obtain a large output voltage. Small deviations in voltage represents the deviation in temperature. It doesn't require external calibration to provide typical accuracies of $\pm 1/40^\circ\text{C}$ at room temperature and $\pm 8/40^\circ\text{C}$ over a full -55 to $+150^\circ\text{C}$ temperature range. LM 35 temperature sensor is low output impedance and has linear output. So, it makes interfacing very easy [7]. Following are some of the features of this temperature sensor:

- Linear + 10-mV/ $^\circ\text{C}$ Scale Factor
- 0.5°C Ensured Accuracy (at 25°C)
- Rated for Full -55°C to 150°C Range
- Suitable for Remote Applications
- Low-Cost Due to Wafer-Level Trimming

- Operates From 4 V to 30 V
- Less Than 60- μA Current Drain
- Low Self-Heating, 0.08°C in Still Air
- Non-Linearity Only $\pm 1/4^\circ\text{C}$ Typical
- Low-Impedance Output, 0.1Ω for 1-mA Load

Since this sensor has many more advantages, so we are using this sensor in our design. The sensor is mounted on the winding of the induction motor and keeps sending voltages to the analog input pins of PLC. The conversion from voltage to temperature can be achieved through simple conversion which states that if there is 10mV at output of LM35, it means 1°C is the temperature and this relation is linearly proportional. The PLC monitors the output of LM35 and if temperature goes beyond the set limit, the motor will be tripped.

V. HARWARE SETUP FOR COMBINED PROTECTION

The protection scheme has been tested on many different induction motors available in laboratory. One of the induction motors parameters are shown in table 1.

TABLE I
PARAMETERS OF AN INDUCTION MOTOR

Parameters	Ratings
Model Number	SN 1500852 12
Delta/Star Voltage	220/380 V
Delta/Star Current	4.3/2.5 A
Power	1.1 kW
Power Factor	0.8
RPM	2870/min
Frequency	50 Hz

The overall combined protection scheme includes all the three protections discussed above in one circuitry as shown in Fig. 3.

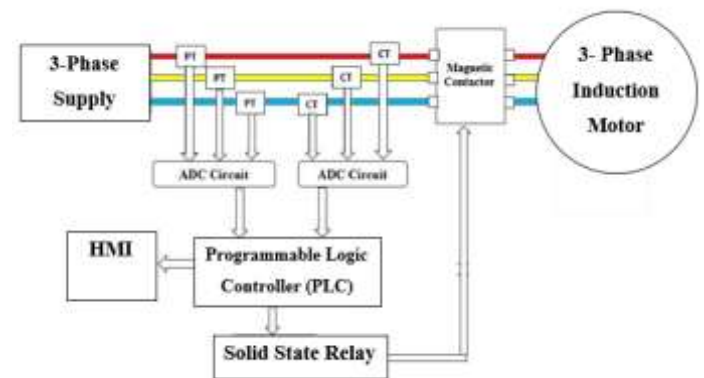


Fig 3. Hardware circuitry diagram of complete protection scheme

The logic for ladder programming is depicted in the form of flow chart in fig. 4. Initially, the motor is started by an operator by pushing the three phase supply button. Since PLC is turned on before the start of induction motor to get to know the supplied voltage, currents and temperature. If all the parameters are normal, the motor will safely start. However, in abnormal condition opposite is true. PLC will continuously monitor and compare all the parameters against the set ones, and will take action as:

- i) If supplied voltages are less than 235 V, PLC gives high output as this is considered as normal state and motor will run.
- ii) If supplied voltages are greater than 235 V, PLC will detect this as overvoltage condition and will trip motor.
- iii) If supplied voltages are less than 180 V, the PLC will detect this voltage as under voltage and will trip the motor.
- iv) Similarly, for winding temperature, if temperature exceeds 40 °C, this will be considered as over temperature condition and PLC will trip the motor.

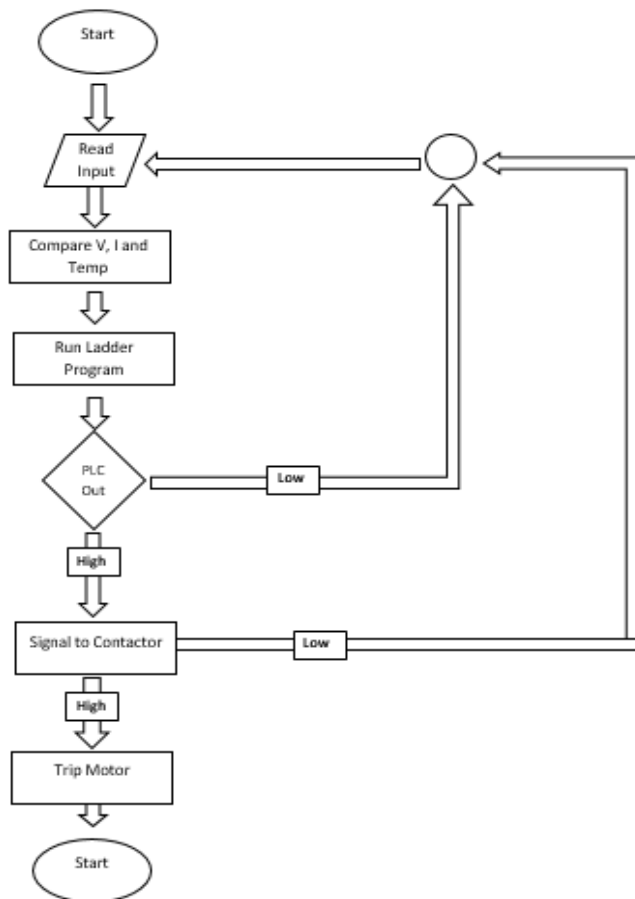


Fig.4. Flowchart of the PLC Logic Program

The normal and abnormal parameters all are shown in Table. II.

TABLE II
TYPES OF FAULTS WITH THEIR NORMAL & ABNORMAL MAGNITUDES

Fault Type	Normal	Abnormal
Over Voltage	220 V	235 V
Under Voltage	220 V	190 V
Over Temperature	27 °C	40 °C

VI. RESULTS AND DISCUSSIONS

The test was performed on the induction motors available in laboratory. Several induction motors with different rated values were taken and the protection scheme was applied on them individually. PLC as a controller kept on monitoring the different parameters. Initially, the test was performed on 1.1kW, 220/380 (Delta/Star), 50Hz induction motor. The normal and abnormal conditions in PLC were set as shown in Table. II. The induction motor was operated from its normal to abnormal condition magnitudes. The voltage magnitude was controlled through the variable power supply available in the laboratory. However, the winding temperature was normal, but abnormal condition was introduced by giving the temperature externally to the LM35 sensor. Following results were achieved as a conclusion of this test:

A. Over Voltage Condition

The normal voltages of the induction motor are set as 220V while the abnormal voltage are set as 235V. Under normal condition, three phase variable power supply is positioned at 220V, the position of relay and magnetic is on as per the logic defined in the ladder programming. The HMI also indicates no any over voltage condition and thus turns indicator green in color as shown in Fig. 5. However Fig.6 shows the results of over voltage condition in graphical view. In normal condition, three phase supply is connected with the motor and motor keeps on running properly on its rated speed at this voltage level.

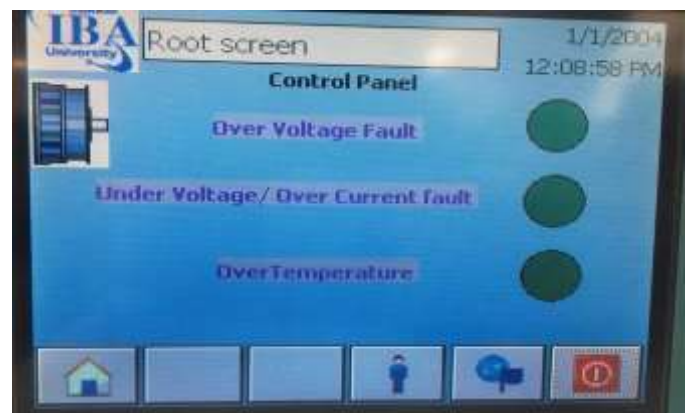


Fig.5. HMI Showing Motor Normal Condition

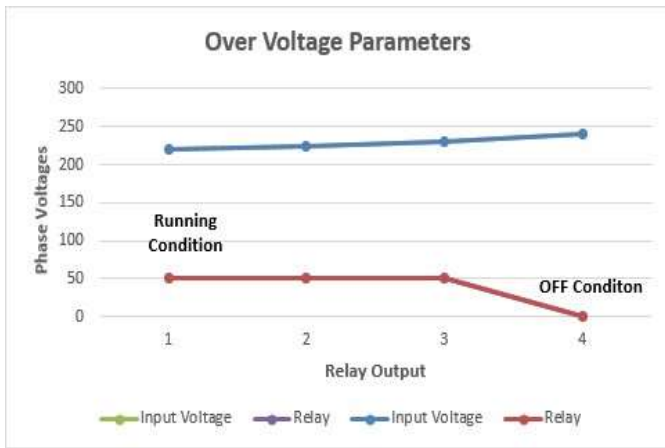


Fig.6. Relay Status during normal & overvoltage condition

However, abnormal condition is introduced when voltage level is increased from its normal value and is set to 235 V. As knob of three phase power supply is positioned at 235V, the ladder programming considers this voltage as abnormal condition and so the output pin of PLC goes high, which again sends signal to solid state relay, which afterwards trips the motor through magnetic contactor. The HMI displays the result by turning green light into red as shown in Fig.7.

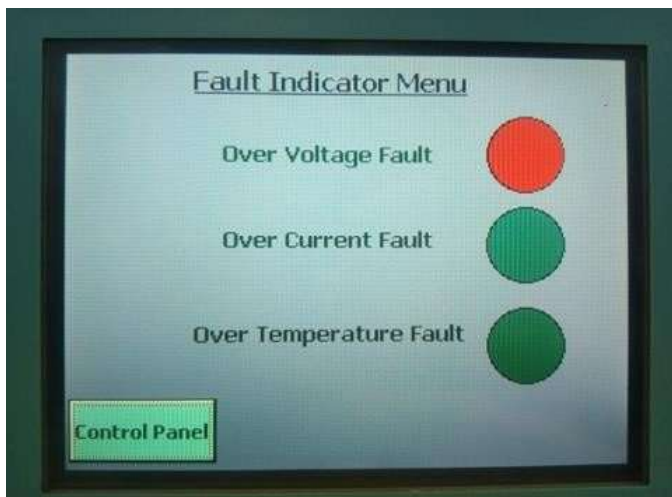


Fig.7. HMI Showing Motor Over Voltage Condition

B. Over Temperature Condition

Under normal condition, the induction motor will be running safely. However, during abnormal condition LM35 generates higher range of voltages (in mV) that are connected to the analog input pin of the PLC. As, the voltage level goes higher than the defined range (the safe temperature), the output of PLC goes high and thus relay is activated and which further sends signal to magnetic contactor and trips the motor. Fig.8. shows the over temperature condition, while fig.9.shows both normal and abnormal temperature state of induction motor.

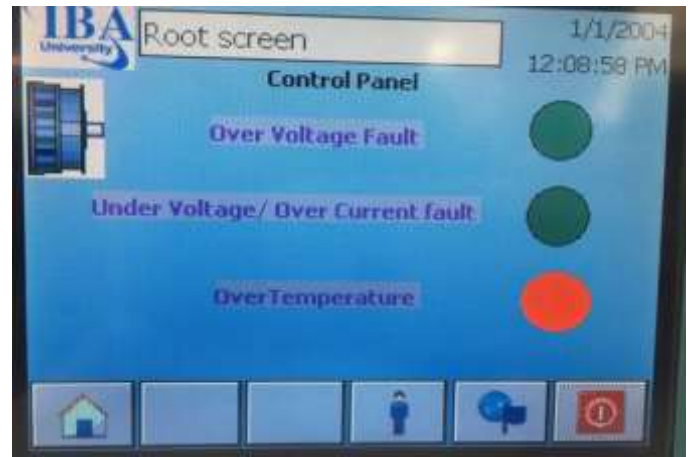


Fig.8. HMI Showing Motor Over Temperature Condition

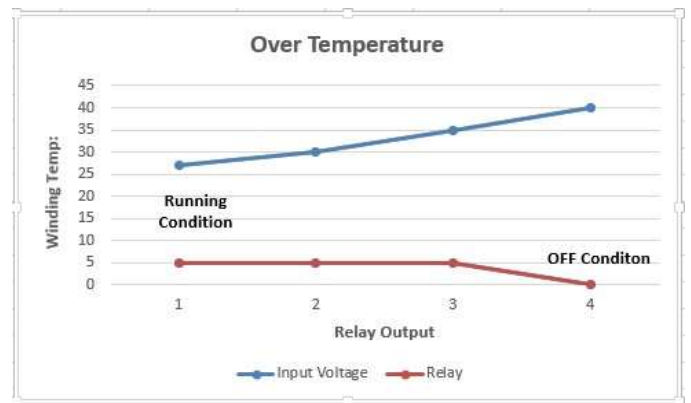


Fig.9. HMI Showing Motor Over Temperature Condition

C. Under Voltage Condition

Under voltage condition or over current occurs when voltages drops below the standard or normal operating voltage level which causes overflow of current in induction motor and electrical stress on the winding results. Since the normal voltage of the induction motor is 220V, whenever this voltage level drops very low, i-e: 190 V, then this voltage level is considered as the under voltage or over current condition. Fig.10. shows the HMI in which under voltage condition is shown.



Fig.10. HMI Showing Motor Under Voltage Condition

Fig.11. shows the normal and under voltage condition of induction motor in graphical view.

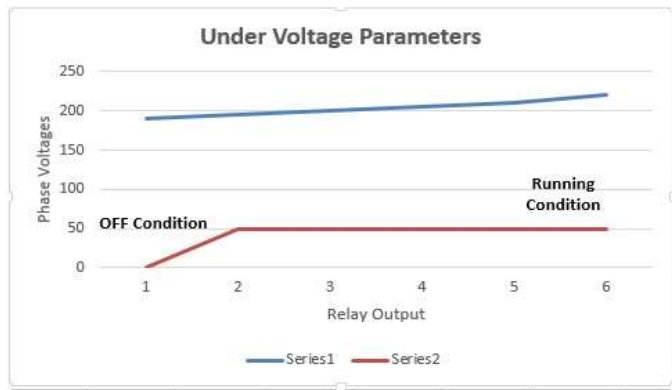


Fig.11. HMI Showing Motor Under Voltage Condition

IX. PLC LADDER PROGRAMMING

Fig.12 shows the ladder logic which essentially uses AND operation for all the three desirable conditions of Temperature, Voltage and Current in terms of Under Voltage. The values indicated in Table 2 are set in the program along with a timer to wait for 2 seconds before the circuit is interrupted.

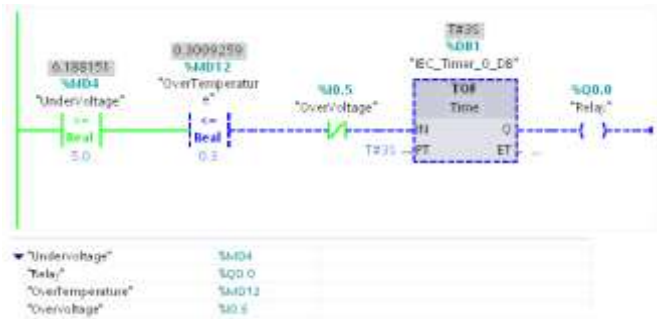


Fig.12. Ladder Logic in TIA Portal

X. CONCLUSION

This paper attempted to present the cost-effective and real time monitoring and protection scheme for industrial three phase induction motors against overcurrent, overvoltage and over-temperature conditions through programmable logic controller. The use of human machine interface is also attempted to bring into application while the need of recloser is mitigated at the same time.

XI. ACKNOWLEDGMENT

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